

Individual Assignment – RFM Response Model

Objective

- The objective of individual assignment 1 is to (1) logistic regression models in customer retention decisions, and (2) practice a few data clean routines.

Zero tolerance policy

- Violation of honor codes will result in a zero on the assignment. Individual assignment must be prepared individually.

Submission Guidelines

- Submit your assignment on Canvas in pdf format.
- Your submission must include:
 1. A single pdf document containing all figures and plots.
 2. You are not required to write a report; focus on presenting the correct figures/plots.
- Name your file as “firstname_lastname.pdf”

Grading

This assignment accounts for **5% of your final grade**. Your grade will be based on:

- **Accuracy:** Whether the figures and plots are correct.
- **Completeness:** Submission of all required figures and answers to each question.

Instructions

1. Read the provided description of the “Book Transaction” dataset.
2. Download the file “**book_transaction.csv**” from Canvas.
3. Answer the assigned questions using the dataset. Each question involves several figures. Generate the required figures and plots for each question and include them directly in the pdf document.

Case Assignment: Book Transactions

The company is an online book seller. We have transaction data for a cohort of 23,570 customers who made their first purchase in the first quarter of 2017. Each record in this file, 69,659 in total, comprises four fields: the customer's ID, the date of the transaction, the number of books purchased, and the dollar value of the transaction.

We will split the data into two time frames and try to predict behavior in period 2 using the transactions from period 1. To do so we will use RFM (recency/frequency/monetary value) analysis and attempt to predict whether a given customer made **at least one purchase** in time period 2. This is similar to assessing whether a customer was retained, though in a non-contractual setting.

Please answer the following questions.

1. Data Restructuring:

Step 1: Split the data into a calibration sample comprising all records on or before 9-30-17, and a validation sample comprising all records after 9-30-17.

Step 2: For both samples restructure the data so that there is one row per customer with the following columns: "ID", "frequency" (number of purchases), "monetary" (average dollar value of purchases), "last.purchase" (the date of the customer's most recent purchase)

Step 3: For both samples, create a column called "recency" which is a numeric variable that represents how many days since the most recent purchase. A good way to do this is to use the last date in the whole sample as an end date. So, a purchase on that date would have a recency score of 0.

Step 4: Merge the calibration and validation samples into one data frame with one row per customer ID.

Step 5: Create a column that takes value 1 if the customer was retained (i.e. if the customer made at least one purchase in the second period) and a 0 if not.

Now sort your new data by ID, and print out the first 10 rows of your newly constructed data in your report.

2. Predicting retention with Logistic regression: predict retention in period 2 using RFM in period 1. Plot out the ROC curve for the logistic regression model. What is the AUC (area under the curve)?

3. Decile analysis: For recency.x, frequency.x, and monetary.x in the restructured data, create new columns that place each customer into a decile based on how they rank. Calculate and plot

the percent retention in each decile of recency, each decile of frequency, and each decile of monetary. Do these variables seem to make any difference to retention?

4. Response models are commonly evaluated in terms of their selectivity, or their ability to sort customers according to their response potential. Based on the estimated logistic regression model,

a. Can you assess the customer selectivity ability of the variable “Recency.x” by plotting out the cumulative lift chart? What about the selectivity ability of the variable “Frequency.x”? What about the selectivity ability of the variable “Monetary.x”? What about the selectivity ability using the predicted retention probability of the logistic regression model?

b. Can you assess the customer selectivity ability of the variable “Recency.x” by plotting out the gain chart? Plot out the gain chart using variables, “Frequency.x”, “Monetary.x”, and the predicted retention probability of the logistic regression model, respectively.